

IF 45 I

Advance Web Programming

Meet6 - Connecting Node.js to MySQL Database

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Meet 6

Objective:

Sub-CLO0713 - Students are able to implement various types of libraries and database to create web application using Node JS (C3)

OUTLINE

- *MySQL Database Node.js driver*
- *Inserting data into MySQL Database*

MySQL Database Node.js driver

Driver

A driver is a library that allows Node.js to communicate with MySQL.

In Node.js, examples of popular drivers for MySQL are:

- mysql (older, but still widely used)
- mysql2 (newer, faster, supports Promises)

Preparing connection

Because we'll use mysql2 for connection with mysql database, before type connection in nodeJS, you should install your mysql2 with this method.

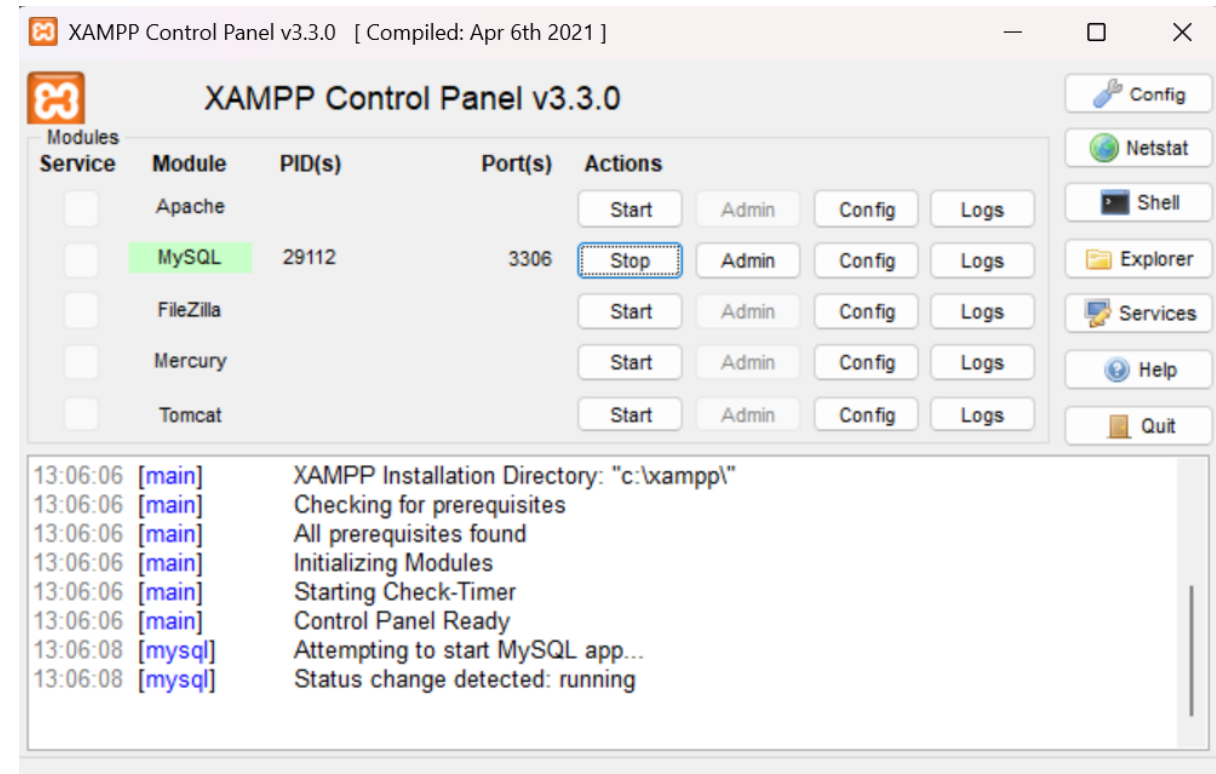
1. Open your terminal and type “npm install mysql2”
2. Or you can use “npm install mysql2 --save”

Or if you want to use old drive “mysql”, you should install mysql with this method:

1. Open your terminal and type “npm install mysql”

Testing connection with mysql database (I)

1. Turnon your mysql from XAMPP
2. If you want to use phpMyAdmin, you can also turnon “Apache” server



Testing connection with mysql database (2)

1. Create new file, with filename, "koneksi.js"
2. After mysql turnon, try to code in your app.js like the sample beside.
3. Don't forget, you need to export variable "connection" with "module.exports"

Note: Make sure that you have created "dbkampus" from previous meeting.

```
PS E:\UMN\IF451 - Ad  
Connected to MySQL!
```

```
const mysql = require("mysql2");

const connection = mysql.createConnection({
  host: 'localhost',
  user: 'root',
  password: '',
  database: 'dbkampus'
});

connection.connect(err => {
  if (err) {
    console.log(err);
  } else {
    console.log("Connected to MySQL!");
  }
});

module.exports = connection;
```


Call other files and Select Query

To call other files from your main files, you can use “require” function.

1. Create your main file, with filename, “app.js”
2. To call “koneksi.js” files, you just only this single line script

```
require("./koneksi.js");
```

3. After that, write the code beside, to show data from table “ms_mahasiswa”

```
const kon = require("./koneksi.js");

kon.query("SELECT * FROM
ms_mahasiswa", (err, results) => {
  if (err) {
    throw err;
  } else {
    console.log(results);
  }
});

kon.end();
```

Call parameter results

To call all parameter from result, you can change or add this code like the sample besides

```
const kon = require("./koneksi.js");

const q = "SELECT * FROM ms_mahasiswa";
kon.query(q, (err, results) => {
  if(err) {
    console.log(err);
  } else {
    results.forEach(elm => {
      console.log(elm.nim);
      console.log(elm.nama);
      console.log(elm.jurusan);
      console.log(elm.alamat);
      console.log(elm.telepon);
      console.log("-----");
    });
  }
});

kon.end();
```

Website form

To show the results in the browser,
change your code like the sample besides

```
const http = require("http");
const kon = require("../koneksi.js");

const server = http.createServer((req, res) => {
  res.writeHead(200, { "Content-Type": "text/html" });

  const q = "SELECT * FROM ms_mahasiswa";
  kon.query(q, (err, results) => {
    if(err) {
      console.log(err);
    } else {
      results.forEach(elm => {
        res.write(elm.nim + "<br>");
        res.write(elm.nama + "<br>");
        res.write(elm.jurusan + "<br>");
        res.write(elm.alamat + "<br>");
        res.write(elm.telepon + "<br>");
        res.write("<hr>");
      });
    }
    res.end();
  });
});

server.listen(3000, () => {
  console.log("Server berjalan di http://localhost:3000");
});
```

Inserting data into MySQL Database

Input via terminal

When you want to input via terminal, first you need to install “readline-sync”

Type this command in your terminal

“npm install mysql2 readline-sync”

And

“npm init -y”

Import Library readline and show data

To import readline library and show data from ms_mahasiswa, please add this sample code to your **app.js**

```
const kon = require("./koneksi.js");
const readline = require("readline-sync");

const server = http.createServer((req, res) => {
  res.writeHead(200, { "Content-Type": "text/html" });

  const nim = readline.question("Masukkan NIM: ");
  const nama = readline.question("Masukkan Nama: ");
  const jurusan = readline.question("Masukkan Jurusan: ");
  const alamat = readline.question("Masukkan Alamat: ");
  const telepon = readline.question("Masukkan Telepon: ");

  // Query Insert
  const sql = "INSERT INTO ms_mahasiswa (nim, nama, jurusan, alamat, telepon)
VALUES (?, ?, ?, ?, ?)";
  kon.query(sql, [nim, nama, jurusan, alamat, telepon], (err, result) => {
    if (err) {
      console.error("Gagal insert:", err);
    } else {
      console.log("Data berhasil ditambahkan, ID:", result.insertId);
    }
  });

  const q = "SELECT * FROM ms_mahasiswa";
  kon.query(q, (err, results) => {
```

Input and Result Insert Data

```
Server berjalan di http://localhost:3000  
Connected to MySQL!  
Masukkan NIM: 3456  
Masukkan Nama: Yanto  
Masukkan Jurusan: DKV  
Masukkan Alamat: Dimana aja  
Masukkan Telepon: 12323123  
Data berhasil ditambahkan, ID: 0
```

localhost:3000	
111	Wirawan
	Informatika
	Gading Serpong
	0123456
222	Budi
	Sistem Informasi
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	02231
3456	Yanto
	DKV
	Dimana aja
	12323123

Input via HTML

First create html file, with filename:
“meet6_sample_input.html” and write the code below

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Form Input Mahasiswa</title>
</head>
<body>
  <form method="POST" action="http://localhost:3000">
    NIM:<input type="text" name="txtNim" required><br>
    Nama:<input type="text" name="txtNama" ><br>
    Jurusan: <input type="text" name="txtJurusan" ><br>
    Alamat: <input type="text" name="txtAlamat" ><br>
    Telepon: <input type="text" name="txtTelp" ><br>

    <input type="submit" value="simpan">
  </form>
</body>
</html>
```


Call HTML file with fs function

To call html file, you need fs function.

Create new file, with name “app.js”
Write the code beside.

And run in terminal

“node app.js”

```
const http = require("http");
const fs = require("fs");

const server = http.createServer((req, res) => {
  if (req.url === "/form") {
    console.log("called...!!! " + req.url);
    fs.readFile("meet6_sample_input.html", (err, data) => {
      if(err) {
        console.log("gagal panggil data html");
      } else {
        res.writeHead(200, { "Content-Type": "text/html" });
        res.write(data);
        res.end();
      }
    });
  }
});

server.listen(3000, () => {
  console.log("Server berjalan di http://localhost:3000");
});
```

Sample code to Get Post data

```
if (req.method === "POST" && req.url === "/") {  
  let body = "";  
  
  req.on("data", chunk => {  
    body += chunk.toString();  
  });  
  
  // setelah semua data terkirim  
  req.on("end", () => {  
    const formData = querystring.parse(body);  
    console.log(formData);  
  });  
}
```

Sample Full Code (I)

```
const http = require("http");
const fs = require("fs");
const querystring = require("querystring");
const kon = require("./koneksi.js");

function tampil_data(res) {
  console.log("tampil data called");
  const q = "SELECT * FROM ms_mahasiswa";
  kon.query(q, (err, results) => {
    if(err) {
      console.log(err);
    } else {
      results.forEach(elm => {
        res.write(elm.nim + "<br>");
        res.write(elm.nama + "<br>");
        res.write(elm.jurusan + "<br>");
        res.write(elm.alamat + "<br>");
        res.write(elm.telepon + "<br>");
        res.write("<hr>");
      });
    }
    res.end();
  });
}
```

Sample Full Code (2)

```
const server = http.createServer((req, res) => {  
  
  if (req.url === "/form") {  
    console.log("called...!!! " + req.url);  
    fs.readFile("meet6_sample_input.html", (err, data) => {  
      if(err) {  
        console.log("gagal panggil data html");  
      } else {  
        res.writeHead(200, { "Content-Type": "text/html" });  
        res.write(data);  
        res.write("<hr>");  
        tampil_data(res);  
      }  
    });  
  }  
});  
}
```

Sample Full Code (3)

```
if (req.method === "POST" && req.url === "/") {
  let body = "";

  req.on("data", chunk => {
    body += chunk.toString();
  });

  // setelah semua data terkirim
  req.on("end", () => {
    const formData = querystring.parse(body);
    console.log(formData);

    // Query Insert
    const sql = "INSERT INTO ms_mahasiswa (nim, nama, jurusan, alamat, telepon) VALUES (?, ?, ?, ?, ?)";
    kon.query(sql, [formData.txtNim, formData.txtNama, formData.txtJurusan, formData.txtAlamat, formData.txtTelp], (err,
result) => {
      if (err) {
        console.error("Gagal insert:", err);
      } else {
        console.log("Data berhasil ditambahkan, ID:", result.insertId);
        res.writeHead(302, {'Location': 'http://localhost:3000/form'});
        res.end();
      }
    });
  });
}
```

Sample Full Code (4) and result

```
server.listen(3000, () => {  
  console.log("Server berjalan di http://localhost:3000");  
});
```

localhost:3000/form

NIM:

Nama:

Jurusan:

Alamat:

Telepon:

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NEXT WEEK'S OUTLINE

- CRUD Operations in MySQL Database

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Visi

Menjadi Program Studi Strata Satu Informatika **unggulan** yang menghasilkan lulusan **berwawasan internasional** yang **kompeten** di bidang Ilmu Komputer (*Computer Science*), **berjiwa wirausaha** dan **berbudi pekerti luhur**.



Misi

1. Menyelenggarakan pembelajaran dengan teknologi dan kurikulum terbaik serta didukung tenaga pengajar profesional.
2. Melaksanakan kegiatan penelitian di bidang Informatika untuk memajukan ilmu dan teknologi Informatika.
3. Melaksanakan kegiatan pengabdian kepada masyarakat berbasis ilmu dan teknologi Informatika dalam rangka mengamalkan ilmu dan teknologi Informatika.